

Java + DSA Learning Guide

This guide is designed for students and aspiring software engineers who want to master Data Structures and Algorithms (DSA) using Java for coding interviews and placements.

1. Java Fundamentals for DSA

- Variables and Data Types
- Loops and Conditional Statements
- Methods and Functions
- Arrays and Strings
- Object-Oriented Programming Basics

2. Time and Space Complexity

- Big-O Notation
- Best, Average, Worst Case
- How to Analyze Complexity
- Optimization Techniques

3. Arrays and Strings

- Traversal
- Prefix Sum
- Sliding Window
- Two Pointer Technique
- String Manipulation

4. Linked Lists

- Singly Linked List
- Doubly Linked List
- Circular Linked List
- Reversal and Cycle Detection

5. Stack and Queue

- Stack Operations

- Queue Operations
- Deque
- Applications of Stack and Queue

6. Trees and Binary Trees

- Tree Traversals
- Binary Search Tree
- Heap
- Priority Queue

7. Graphs

- Graph Representation
- BFS
- DFS
- Shortest Path Algorithms

8. Algorithms

- Sorting Algorithms
- Searching Algorithms
- Recursion
- Backtracking
- Dynamic Programming

9. Interview Preparation

- LeetCode Practice
- Problem Solving Strategy
- Mock Interviews
- Competitive Programming Basics

10. Projects and Practice

- Library Management System
- Banking System
- Student Management System
- Mini Coding Challenges

Recommended 6-Month DSA Roadmap

Month	Focus Area
1	Java Basics + Arrays + Strings
2	Linked Lists + Stack + Queue
3	Trees + Heap + BST
4	Graphs + Recursion
5	Dynamic Programming + Backtracking
6	LeetCode + Mock Interviews + Revision

Recommended Platforms:

- LeetCode
- HackerRank
- GeeksforGeeks
- CodeStudio
- GitHub for Project Uploads

Consistency is the key to mastering DSA. Practice daily, understand problem-solving patterns, and build strong logic using Java to prepare for product-based company interviews.